

Date:

Part A

Question 1

Declare i, N AS INTEGERS

OUTPUT "Enter numbers you want to print"

INPUT N

$i \leftarrow N$

FOR COUNT $\leftarrow 0$ TO N

OUTPUT i

$i \leftarrow i - 1$

END FOR

Question 2

DECLARE Year : INTEGER

OUTPUT "Enter Year"

INPUT Year

IF (Year MOD $= 0$ AND Year MOD $\neq 0$) OR

(Year MOD 400 $= 0$)

PRINT Year, "is a leap year"

ELSE

PRINT Year, "is not a leap year"

END IF

Question 3

```

DECLARE n : INTEGER
INPUT n
IF (n > 0 AND n < 10) THEN
    OUTPUT "Is a single digit number"
ELSEIF (n > 9 and n < 100) THEN
    OUTPUT "It is a two digit number"
ELSE
    OUTPUT "It is three digit or more"

```

Question 4

```

DECLARE i, j : INTEGERS
FOR i ← 1 TO 10
    IF i MOD 3 ≠ 0 THEN
        FOR j ← 1 TO 10
            OUTPUT i, " x ", j, " = ", i * j
        END FOR
    ELSE
        NEXT
    END FOR

```


Question 5

DECLARE n : INTEGER

INPUT num

WHILE num $\geq$ 10

sum $\leftarrow$ 0

WHILE $n > 0$

digit $\leftarrow$ $n \bmod 10$

sum $\leftarrow$ sum + digit

$n \leftarrow n \text{ DIV } 10$

ENDWHILE

$n \leftarrow$ sum

ENDWHILE

OUTPUT n